

Miracle Chimamanda Ndubuisi

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PROFILE

Offensive Security Engineer with hands-on experience in penetration testing, red team operations, SIEM monitoring, and incident response across mobile, web, API, and cloud infrastructure. Currently focused on the intersection of offensive security and AI - how AI systems can be attacked, how they change the threat landscape, and how offensive techniques can be automated with machine learning. Seeking a PhD to explore AI security and offensive security research, building on practical experience breaking into production fintech systems, 40+ vulnerabilities discovered across professional engagements, and 3 published research papers.

RESEARCH INTEREST

- Automating penetration testing and vulnerability discovery with AI/ML
- Security risks in mobile and web apps that integrate AI features
- How large language models can be exploited or misused in security contexts
- Using machine learning to improve threat detection and intrusion detection

EDUCATION

Alex Ekwueme Federal University, Ebonyi State, Nigeria

Sept 2017 – Dec 2021

Bachelor of Science (Ed) – Physics, *First Class Honours*

- Thesis: Awareness of Artificial Intelligence among Pre-service Teachers in Ebonyi State
- Original survey design, quantitative data collection (200+ participants), and statistical analysis (SPSS), culminating in a peer-reviewed publication.

Relevant Coursework: Computer Science I & II, Elementary Mathematics I & II, Elementary Mechanics I & II, Mathematical Methods in Physics I & II, Electronics, Electronic Measurement & Instrumentation, Electric Circuits & Electronics, Electromagnetism & Atomic Physics, Quantum Physics I & II, Thermal Physics, Research Methods & Statistics, Research Project.

HONOURS & DISTINCTIONS

- First-Class Honours, highest distinction awarded, top of graduating class in the Department of Physics and Science Education.

- Published peer-reviewed research as an undergraduate, a distinction achieved by fewer than 5% of students in the department.
- Admitted to the 3-Million Technical Talents (3MTT) programme, a competitive Federal Government of Nigeria initiative for top technology professionals.

PUBLICATIONS

First-Author Work [1] **Ndubuisi, M. C.**, Abanum, C., & Adigwu, D. U. (in review). AI: A Danger to an Illiterate Society. Knowledge and Information System. – Under review at Springer. Examines societal risks of AI in low-digital-literacy populations; proposes governance frameworks.

Co-Authored Work[2] Chiago, C. A., **Ndubuisi, M. C.**, et al. (2024). Evaluating the Impact of AI on Medical Physics: A Mini Review. REJOST, 4(5), 22–43. – Systematic review of AI in diagnostics, treatment planning, and medical imaging (40+ sources).

[3] Agbo, J. N., Nmadu, J., **Ndubuisi, M. C.**, et al. (2022). Awareness, Willingness, and Readiness of Pre-Service Teachers to Deploy AI. AJSTME, 8(1), 25–31. – Quantitative study with original validated instruments; findings informed curriculum reform recommendations.

RESEARCH EXPERIENCE

Security Research & Engineering – Threat Detection & Network Security Jan 2026 – Present Hizo Technology Limited

- Perform daily security operations as a Security Engineer and Junior Mobile Application Pentester across web apps, mobile apps, APIs, networks, and cloud infrastructure.
- Analyse network traffic patterns and security alert logs to identify gaps and weaknesses in current detection systems.
- Exploring how machine learning techniques could improve threat detection based on patterns observed through daily hands-on security work.
- Document security findings and research insights to support continuous improvement of the organisation's security posture.

Undergraduate Thesis Research Sept 2020 – Feb 2021 Alex Ekwueme Federal University
Department of Science Education

- Led original quantitative research on AI awareness among 200+ pre-service teachers; designed and validated a survey instrument (Cronbach's $\alpha > 0.80$); applied descriptive and inferential statistics (SPSS).
- Findings contributed to a peer-reviewed publication (Agbo et al., 2022); thesis defence commended for methodological rigour.

Capstone – Secure Application Deployment with Automated Monitoring June 2025 Digital Witch Academy | Cloud Security & DevOps Programme

- Deployed a containerised application (Docker/Kubernetes) with RBAC, namespace isolation, and network policies; integrated Datadog APM for real-time anomaly alerting.

Laboratory Research Assistant (Part-time) Nov 2020 – Mar 2021 Alex Ekwueme Federal University | Department of Physics

- Supported experimental physics research (optics, electronics, mechanics); co-developed protocols; mentored 15+ junior students on experimental design and scientific writing.

PROFESSIONAL EXPERIENCE

Security Engineer & Junior Mobile Application Pentester Jan 2026 – Present Hizo Technology Limited, Asaba, Nigeria

- Perform mobile application penetration testing (Android/iOS): identifying vulnerabilities in client-side logic, insecure data storage, API communication, authentication flows, and certificate pinning.
- Built and maintained a real-time alerting and monitoring dashboard to detect and respond to suspicious activity and security incidents.
- Conduct offensive and defensive security operations: penetration testing, vulnerability assessments, threat hunting, incident response, and SIEM monitoring across web apps, mobile apps, APIs, networks, and cloud infrastructure.
- Execute red-team operations simulating adversarial TTPs mapped to MITRE ATT&CK; develop custom exploit PoCs to validate severity and drive remediation.
- Lead incident investigations: triage, digital evidence preservation, impact assessment, containment, and post-incident reporting.
- Produce technical reports with CVSS scoring, root-cause analysis, and remediation guidance for technical and executive audiences.
- Monitor SIEM dashboards, network captures, and UBA to detect anomalous activity and IOCs in real time.
- Collaborate with dev/DevOps teams on OWASP ASVS implementation; author security policies, SOPs, and playbooks.

DevOps Engineer Intern July 2025 – Dec 2025 Hizo Technology Limited, Asaba, Nigeria

- Designed CI/CD pipelines (GitHub Actions) with SAST/DAST scanning and multi-environment delivery reducing manual deployment effort.
- Containerised 3+ applications (Docker/Kubernetes); provisioned infrastructure on DigitalOcean via Terraform.
- Configured Cloudflare WAF, DDoS mitigation, and bot management; integrated Datadog APM for real-time monitoring.
- Hardened servers: SSH key-only auth, UFW/iptables, fail2ban, least-privilege IAM, automated patching, TLS.

SELECTED TECHNICAL PROJECTS

Network Security & IDS Research Lab (Personal)**2026 – Present**

- Built a virtualised lab for network security experimentation: traffic capture (Wireshark), vulnerability scanning (Nmap), exploitation (Metasploit), and web app testing (Burp Suite).

CI/CD Optimisation & Dockerisation (Hizo)**Aug 2025**

- Re-engineered legacy workflows into automated CI/CD with Docker; embedded SAST/DAST security gates and container vulnerability scanning.

Cloudflare Hardening for Web Application Security (Hizo)**July 2025**

- Configured WAF rules, rate limiting, IP reputation filtering, and bot management protecting a production SaaS app—zero successful attacks during deployment.

Terraform Infrastructure Deployment on DigitalOcean (Hizo)**July 2025**

- Provisioned production-ready infrastructure via Terraform with VPC networking, firewall rules, SSH key-only access, and encrypted storage.

TEACHING EXPERIENCE**Physics Teacher — Senior Secondary (SS1 & SS2)****Nov 2022 – Dec 2023****Zappa Mixed Secondary School, Asaba, Delta State, Nigeria**

- Delivered physics curriculum to ~120 students; conducted 30+ lab sessions; average scores improved ~15% over two exam cycles.
- Led STEM outreach workshops in underserved communities in Oshimili South LGA.

Mathematics Teacher**Jan 2022 – Sept 2022****Prime Academy School, Enugu State, Nigeria**

- Developed lesson plans and assessments aligned with national standards; used differentiated instruction and peer tutoring.

TECHNICAL SKILLS

Programming: Python (advanced), SQL, JavaScript, Bash/Shell, HTML/CSS, Markdown
Offensive Security: Penetration testing (web, mobile, network, API), red-team ops, exploit dev, MITRE ATT&CK, OWASP Top 10/ASVS/MASTG, Burp Suite, Nmap, Metasploit, Wireshark, Frida, MobSF

AI & ML: Scikit-learn, Pandas, NumPy, Streamlit, LangChain

Defensive Security: SIEM (Splunk, ELK), IDS/IPS, incident response, digital forensics, UBA, Cloudflare WAF/DDoS, fail2ban, UFW/iptables

Cloud & Infra: AWS (certified), GCP, DigitalOcean, Docker, Kubernetes, Terraform, Datadog, CI/CD (GitHub Actions), IaC, TLS

Research: Quantitative/qualitative design, survey development & validation, SPSS, systematic literature review, mixed-methods, Zotero, Overleaf/LaTeX, academic writing

TRAINING/CERTIFICATIONS

AWS Certified Cloud Practitioner	2025
Cloud Security and DevOps, Digital Witch Academy	2025
Cloud Computing, 3-Million Technical Talents (3MTT), Abuja, Nigeria	2024
Information System, Modern States (Online)	2024
English for IT 1& 2, Cisco (Online)	2024
Careers Essential in Cybersecurity, Microsoft & LinkedIn	2024
International Business Machines (IBM) Corporation – Cybersecurity Analyst, Coursera	2023

LANGUAGES

English (Fluent — IELTS 7.0, CEFR C1) | Igbo (Fluent — native) | German (Basic / A1)

LEADERSHIP & COMMUNITY SERVICE

Fire Book Place - Public Relations Officer	2024 – Present
Coordinating outreach and communication for a community-based educational and literary platform.	
Pad a Girl Initiative (Pad a Girl Africa)	2023
Volunteered in advocacy and support programs to empower girls through access to sanitary products and mentorship.	
Online Physics Sessions (Personal Project)	2023
Organized and facilitated virtual physics tutorials for high school students to improve STEM access.	
Free Lesson Classes for Indigent Children (Personal Project)	2020
Provided free science and math lessons for underserved students in the local community.	
Drama Coordinator - Young Anglican Crusaders •	2019–2022
Directed creative programs, mentorship, and youth development activities.	
Course Representative / Faculty Vice President – Alex Ekwueme Federal University	2018–2019
Represented students in academic and faculty matters; coordinated student-led initiatives.	
Vice President, Young Anglican Crusaders	2014 – 2017
Provided leadership for youth fellowship programs and community outreach activities.	
Volunteer Facilitator (Girls' Guild) – Skill Acquisition Training	2018
Supported vocational training workshops to build skills and confidence in young girls	

REFERENCES

Dr. Jovial N. Agbo | Dept. of Science Education, Alex Ekwueme Federal University | Thesis supervisor & research collaborator | agbojovial@gmail.com

Dr. Udoka Nwankwo | Dept. of Physics, Alex Ekwueme Federal University | Academic mentor & faculty advisor | udoka.nwankwo@connect.polyu.hk

Dr. David Udu | Dept. of Science Education, Alex Ekwueme Federal University | Research collaborator | david.udu@funai.edu.ng